PUSHING A FINITE TYPE IDEAL ALONG A QUASI-COMPACT SEPARATED ETALE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 8715-8775, immediately following the statement at 8697-8713.

Editorial note. The Fitting ideal computation is displayed with all three equalities inside the span.

Editorial note. Six cross-chapter pointers (More on Algebra, Descent on Spaces, Limits of Spaces, Morphisms of Spaces) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Fitting ideals and their behaviour under finite locally free morphisms. More on Algebra, lemma more-algebra-lemma-fitting-ideal-basics, cited at proof lines 8770-8772. Computes the Fitting ideal of the pushed module and is the algebraic core of the construction.
- Extension of finite modules over thickenings. Limits of Spaces, lemma spaces-limits-lemma-extend, cited in the proof. Extends the relevant finite module from a closed piece to a neighbourhood, which is how the ideal I' is produced.
- Finiteness properties of closed immersions and compositions. Morphisms of Spaces lemmas on finite presentation of closed immersions, compositions and integral morphisms, cited in the proof. Keeps the constructed ideal of finite type along the auxiliary closed subscheme Z and its finite piece Z'.
- The pass-through-finite addendum. chapter lemma lemma-quasi-finite-separated-pass-through-finite-addendum, cited in the proof. Supplies the finite piece Z' of the auxiliary construction.

Source reference. Original lemma 0874: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7806–7806. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07ZA: focused-seeds/geometry/S9-spaces-push-ideal/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0610: focused-seeds/geometry/S9-spaces-push-ideal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0853: focused-seeds/geometry/S9-spaces-push-ideal/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084Q: focused-seeds/geometry/S9-spaces-push-ideal/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03XQ: focused-seeds/geometry/S9-spaces-push-ideal/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03ZR: focused-seeds/geometry/S9-spaces-push-ideal/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let $\varphi : W \to X$ be a quasi-compact separated étale morphism. Let $U \subset X$ be a quasi-compact open subspace. Let $\mathcal{I} \subset \mathcal{O}_W$ be a finite type quasi-coherent sheaf of ideals such that $V(\mathcal{I}) \cap \varphi^{-1}(U) = \emptyset$. Then there exists a finite type quasi-coherent sheaf of ideals $\mathcal{J} \subset \mathcal{O}_X$ such that*

(1) *$V(\mathcal{J}) \cap U = \emptyset$, and*
(2) *$\varphi^{-1}(\mathcal{J})\mathcal{O}_W = \mathcal{I}\mathcal{I}'$ for some finite type quasi-coherent ideal $\mathcal{I}' \subset \mathcal{O}_W$.*

Proof. Choose a factorization $W \to Y \to X$ where $j : W \to Y$ is a quasi-compact open immersion and $\pi : Y \to X$ is a finite morphism of finite presentation (Lemma 0874). Let $V = j(W) \cup \pi^{-1}(U) \subset Y$. Note that $\mathcal{I}$ on $W \cong j(W)$ and $\mathcal{O}_{\pi^{-1}(U)}$ glue to a finite type quasi-coherent sheaf of ideals $\mathcal{I}_1 \subset \mathcal{O}_V$. By Limits of Spaces, Lemma 0853 there exists a finite type quasi-coherent sheaf of ideals $\mathcal{I}_2 \subset \mathcal{O}_Y$ such that $\mathcal{I}_2|_V = \mathcal{I}_1$. In other words, $\mathcal{I}_2 \subset \mathcal{O}_Y$ is a finite type quasi-coherent sheaf of ideals such that $V(\mathcal{I}_2)$ is disjoint from $\pi^{-1}(U)$ and $j^{-1}\mathcal{I}_2 = \mathcal{I}$. Denote $i : Z \to Y$ the corresponding closed immersion which is of finite presentation (Morphisms of Spaces, Lemma 084Q). In particular the composition $\tau = \pi \circ i : Z \to X$ is finite and of finite presentation (Morphisms of Spaces, Lemmas 03XQ and 03ZR).

Let $\mathcal{F} = \tau_*\mathcal{O}_Z$ which we think of as a quasi-coherent $\mathcal{O}_X$-module. By Descent on Spaces, Lemma 0610 we see that $\mathcal{F}$ is a finitely presented $\mathcal{O}_X$-module. Let $\mathcal{J} = \mathrm{Fit}_0(\mathcal{F})$. (Insert reference to fitting modules on ringed topoi here.) This is a finite type quasi-coherent sheaf of ideals on X (as $\mathcal{F}$ is of finite presentation, see More on Algebra, Lemma 07ZA). Part (1) of the lemma holds because $|\tau|(|Z|) \cap |U| = \emptyset$ by our choice of $\mathcal{I}_2$ and because the 0th Fitting ideal of the trivial module equals the structure sheaf. To prove (2) note that $\varphi^{-1}(\mathcal{J})\mathcal{O}_W = \mathrm{Fit}_0(\varphi^*\mathcal{F})$ because taking Fitting ideals commutes with base change. On the other hand, as $\varphi : W \to X$ is separated and étale we see that $(1, j) : W \to W \times_X Y$ is an open and closed immersion. Hence $W \times_Y Z = V(\mathcal{I}) \amalg Z'$ for some finite and finitely presented

morphism of algebraic spaces $\tau' : Z' \to W$. Thus we see that

$$\begin{aligned}\mathrm{Fit}_0(\varphi^*\mathcal{F}) &= \mathrm{Fit}_0((W \times_Y Z \to W)_*\mathcal{O}_{W\times_Y Z}) \\ &= \mathrm{Fit}_0(\mathcal{O}_W/\mathcal{I}) \cdot \mathrm{Fit}_0(\tau'_*\mathcal{O}_{Z'}) \\ &= \mathcal{I} \cdot \mathrm{Fit}_0(\tau'_*\mathcal{O}_{Z'})\end{aligned}$$

the second equality by More on Algebra, Lemma 07ZA translated in sheaves on ringed topoi. Setting $\mathcal{I}' = \mathrm{Fit}_0(\tau'_*\mathcal{O}_{Z'})$ finishes the proof of the lemma. □